

03 — Household Waste Incineration: The Power with the Highest Hidden Costs

How European waste incineration plants have become disguised as sustainable energy sources, whilst remaining the power generators with the highest hidden costs on the continent.

What this episode covers

- The actual emissions: 2.5–2.9 tonnes of CO₂ per tonne of plastic incinerated (GAIA, Energy Justice, EPA eGRID).
- Per kWh: WIPs emit 2.5 times more CO₂ than a coal-fired power station and 4–5 times more than a natural gas power station.
- The plastic fraction in household waste: 12–15% by mass, yet providing ~80% of the calorific value and ~70% of the CO₂ emissions.
- The legal construct facilitating this: Landfill Directive 1999/31/EG, Waste Framework Directive 2008/98/EG, RED III.
- The European figures: ~50 megatonnes of CO₂/year from WIPs, of which ~35 megatonnes originate from the plastic fraction.
- Why this remains a dogma: ‘indispensable’ for hygiene, volume reduction, district heating, and compliance with EU directives.

This episode is still being elaborated with the installation data of the largest European WIPs (AEB Amsterdam, MVA Rothensee, Spittelau Wenen) and the econometric calculation of the actual kWh-to-CO₂ ratio per facility.